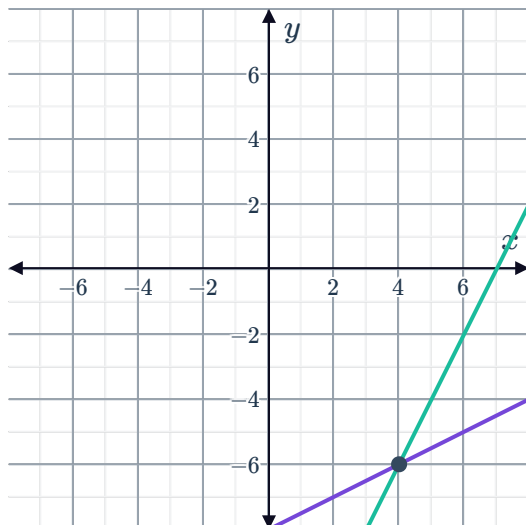


Graphical solutions

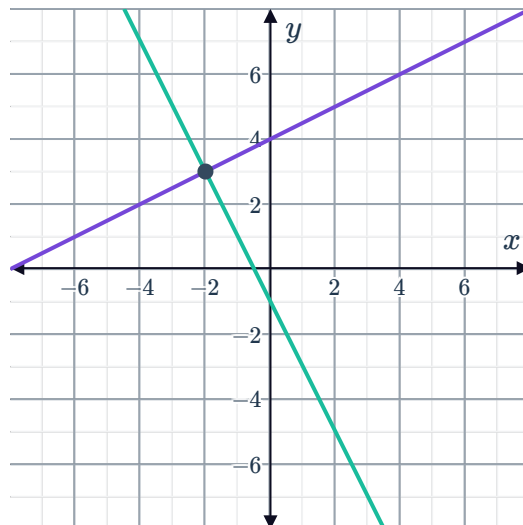


- 1 Each graph below shows a system of two equations. Write down the solution to each system, as a point in the form (x, y) .

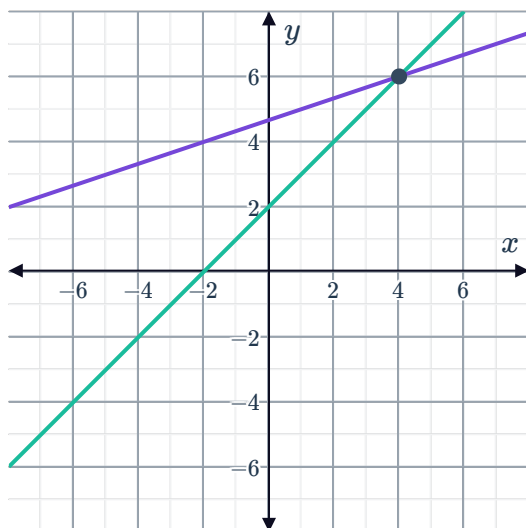
a



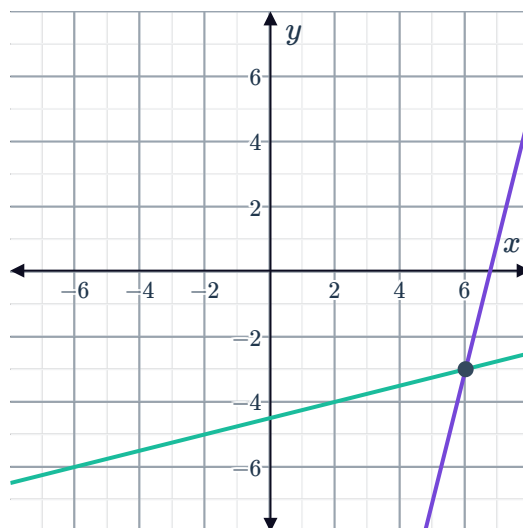
b



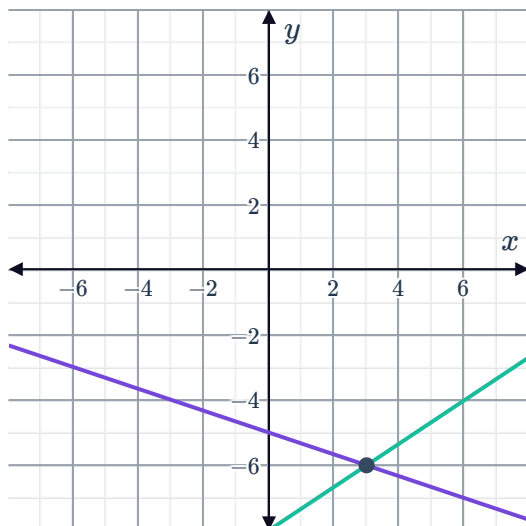
c



d



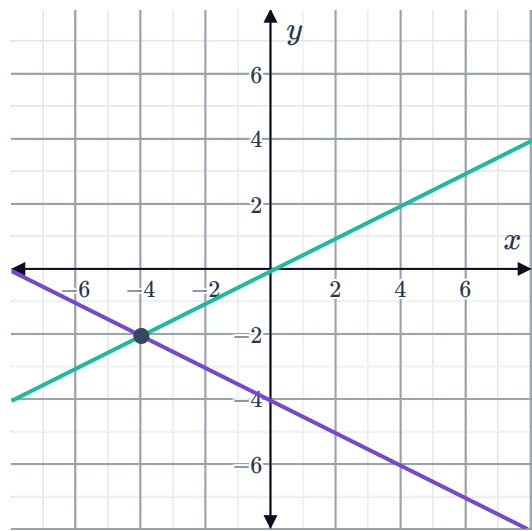
e



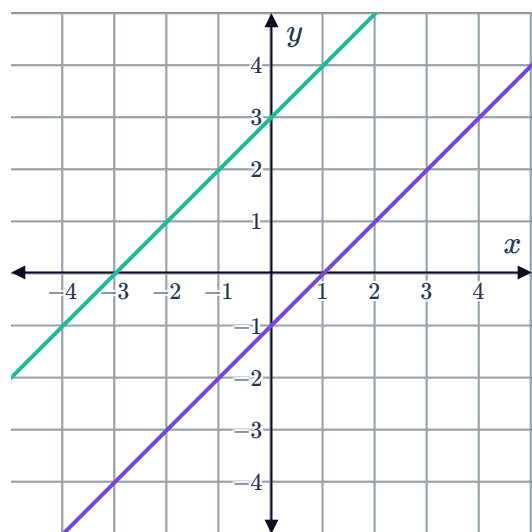
2 Consider the graph showing a system of two equations:



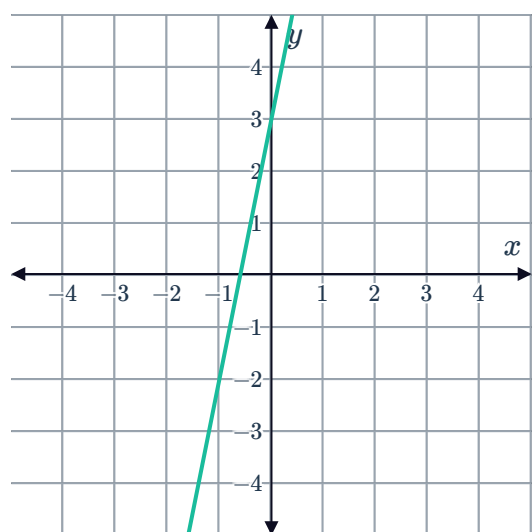
- a How many solutions does this system of equations have?
- b Write down the solution to the system of equations as an ordered pair (x, y) .



3 Consider the graph showing a system of two equations:
How many solutions does this system of equations have?



4 Consider the graph of the equation $y = 5x + 3$:
If a second line $y = mx + b$ intersects this line at the point $(0, 3)$, state the value of b .



5 Consider the following systems of linear equations:



- i Find the x and y -intercepts of the first equation.
- ii Find the x and y -intercepts of the second equation.
- iii Graph the two equations on the same set of axes.
- iv State the coordinate pair (x, y) that satisfies both equations.

a $y = x - 1$
 $y = -2x + 8$

b $y = x + 1$
 $y = 4x + 4$

c $y = x + 3$
 $y = 3x + 3$

d $y = -2x + 2$
 $y = x - 7$

e $y = -x - 1$
 $y = -3x - 3$

f $y = 2x + 2$
 $y = -x + 2$

6 Consider the following linear equations:

$$y = 2x + 2$$

$$y = -2x + 2$$

- a Determine the gradient and y -intercept of the line $y = 2x + 2$.
- b Determine the x and y -intercepts of the line $y = -2x + 2$.
- c Hence sketch the graphs on the same coordinate plane.
- d State the values of x and y which satisfy both equations.

7 Consider the following system of equations: $y = 4x - 3$
 $y = 4 - 3x$

a Complete the table of values for $y = 4x - 3$:

x	-3	-2	-1	0	1	2	3
y							

b Complete the table of values for $y = 4 - 3x$:

x	-3	-2	-1	0	1	2	3
y							

- c Find the gradient of $y = 4x - 3$.
- d Find the gradient of $y = 4 - 3x$.
- e Graph the two equations on the same set of axes.
- f Using your answer from part (e), find the point of intersection of the lines.



$$y = x - 2$$

$$y = -3x - 6$$

- a Find the gradient, and the y -value of the y -intercept, of the line $y = x - 2$.
 - i gradient
 - ii y -value of the y -intercept
- b Find the gradient, and the y -value of the y -intercept, of the line $y = -3x - 6$.
 - i gradient
 - ii y -value of the y -intercept
- c Graph the two equations on the same set of axes.
- d State the coordinate pair (x, y) that satisfies both equations.

9 Consider the following system of equations: $y = -4x - 1$

$$y = -4x - 1$$

$$y = -4x + 2$$

- Graph the two equations on the same set of axes.
- Does there exist a value for x and y that satisfy the following two equations simultaneously?

10 For each of the following systems of equations:

- Graph the two equations on the same set of axes.
- State the values of x and y that satisfy both equations.

a $y = x$
 $y = -1$

b
$$\begin{aligned} 4x - 2y &= 2 \\ -2x + 4y &= 2 \end{aligned}$$